

Frontend Technologies - Session 1 Assignment

Project Requirements: Swift Shop Vendor Dashboard

Project Overview: Develop a responsive, E-commerce Vendor Portal using React 19. This dashboard allows merchants to manage their product catalog, track inventory status, and filter through their offerings dynamically.

1. Technical Stack Requirements

- **Framework:** React 19 (using Vite)
- **Styling:** Tailwind v4 CSS (Utility-first approach)
- **Icons:** Your choice

2. Core Functional Requirements

2.1 Component Architecture (The "Base")

- **Layout Wrapper:** Implement a persistent Navbar for desktop and a responsive Sidebar for mobile layout.
- **Product Card:** Create a reusable UI component to display product details.
- **Stat Tiles:** Create small components to show "Total Products," "Low Stock Alert," and "Total Inventory Value."

2.2 Product Catalog (List & Key Logic)

- **Dynamic Rendering**
 - *Hint: Map through a list of products*
- **Key Integrity:** You must use unique id values for keys.
- **Conditional UI:** If stock is 0: Show an "Out of Stock" overlay.
 - If stock is < 5: Show a "Limited Quantity" badge in orange.
 - If price is > \$500: Display a "Premium" tag.

2.3 Management Form (Controlled Components)

- **Add Product Form:** A form with inputs for Name, Price, Category, and Stock Quantity.
- **State Handling:** Use a **single object state** to manage all form inputs.
- **Validation:** Prevent submission if the name is empty or the price is less than or equal to 0.
- Edit Product
- Delete Product

Hint: You can store data in local storage or Session Storage. Data is not the concern here just it should be stored somewhere

2.4 Search & Filtering (State Logic)

- **Real-time Search:** A search bar that filters the displayed products by name as the user types.
- **Category Filter:** A dropdown menu to filter products by category (e.g., Electronics, Fashion, Home).

3. Technical Constraints

Below are few common points that may help you to maintain code quality

1. **Immutability:** Never mutate state directly. Use the spread operator (...) for updates.
2. **Prop Destructuring:** Props must be destructured within the function parameters.
3. **Declarative Patterns:** Avoid imperative DOM manipulation (no document.querySelector).
4. **Tailwind Responsiveness:** The dashboard must be a 1-column layout on mobile (< 640px) and a 3-column grid on desktop (> 1024px). You are free to choose your dimensions but should be near to real ones

4. Submission Guidelines

- **Format:** GitHub Repository link.
- **Deliverables:**
 - Fully functional Vite/React application.
 - Please commit the code before leaving the office, even if the assignment is incomplete. That is okay. There should be at least one commit from at the end of the day.

Nice to have

- A README.md explains how you handled the "Add Product" state.
- A brief comment on the code explaining why **React Fragments** were used in your components.